Arista Access Node Recovery Procedure for Hardware/Environmental Faults (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss) at Jayanagar Metro Site

Node: JAY-ACC-47 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9725

Description

The occurrence of hardware or environmental faults such as ASIC memory parity errors, line card crashes, or power supply redundancy loss on an Arista access node (e.g., Node ID: JAY-ACC-47) can be attributed to the intricacies of the node's underlying architecture. In an access role, the node is responsible for aggregating traffic from various endpoints and forwarding it towards the distribution or core layers of the network. The Arista EOS (Extensible Operating System) utilizes a modular design, with each line card operating independently, leveraging the node's ASICs for packet processing and forwarding. ASIC memory parity errors indicate data corruption within the ASIC, potentially due to hardware failures or environmental factors such as temperature fluctuations or power supply issues. Line card crashes can result from similar causes, affecting the node's ability to forward traffic. Power supply redundancy loss compromises the node's high availability, risking complete system failure. Given the critical nature of these components in maintaining network uptime and performance, a meticulous and swift response to such failures is essential.

Pre-requisite Checks

- Verify site accessibility and safety for remote hands at Jayanagar Metro Site
- Check the status of redundant links to ensure no single point of failure
- Confirm the availability of necessary tools and spare parts

Potential Root Causes

Cause	Probability
Hardware Failure (ASIC, Line Card, Power Supply)	High
Environmental Factors (Temperature, Humidity, Power Quality)	Medium
Software Bug or Incompatibility	Low

Diagnostic Commands & Expected Outputs

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Command: show hardware forwarding ASIC register, Expected Output: Detailed register values indicating ASIC health, Description: This command isolates issues within the ASIC by displaying its current state, helping to identify any memory parity errors or other hardware malfunctions affecting packet forwarding.
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Command: show linecard, Expected Output: Line card status, including any faults or errors, Description: This command provides insights into the operational status of each line card, including any crashes or errors that could be contributing to the failure domain.

Command: show environment power, Expected Output: Power supply redundancy status and individual power supply health, Description: Verifies the power supply redundancy and health, ensuring that no power-related issues are causing or contributing to the hardware or environmental faults.
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Resolution Steps

Step 1: Step: Replace Faulty Hardware Component, Command: hardware module replace , Description: If a hardware component such as an ASIC, line card, or power supply is found to be faulty, this step involves replacing it with a functional spare, ensuring the node's return to full operational capacity.

Step 2: Step: Apply Environmental Remediations, Command: No specific command; involves adjusting site environmental conditions, Description: If environmental factors are identified as the root cause, this step may involve adjusting the site's temperature, humidity, or ensuring a stable power supply to prevent future occurrences.

Step 3: Step: Update or Rollback EOS Version if Necessary, Command: boot system flash , Description: If a software bug is suspected, updating or rolling back the EOS version to a known stable release may resolve the issue, ensuring compatibility and stability of the node's operating system.

Post-Check Verification

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Command: show hardware forwarding ASIC register, Expected Output: Registers indicating healthy ASIC operation, Description: Re-executes the diagnostic command to verify that the ASIC health has been restored, confirming the resolution of memory parity errors.

Command: show linecard, Expected Output: All line cards operational with no errors, Description: Confirms that all line cards are functioning correctly, with no signs of crashes or other issues that could impact traffic forwarding.

Command: show environment power, Expected Output: Power supplies in redundant configuration and healthy, Description: Verifies that power supply redundancy has been restored and all power supplies are operational, ensuring the node's high availability.
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Roll-back Procedures

- Step: Revert to Previous Hardware Configuration, Command: hardware module replace , Description: If replacing a hardware component introduces new issues or does not resolve the problem, this step involves reinstalling the original component to return the node to its previous state.

- Step: Revert Environmental Adjustments, Command: No specific command; reverses environmental adjustments made, Description: If environmental adjustments were made and found not to resolve the issue, this step reverses those changes to avoid any unintended consequences on the site's conditions.

- Step: Revert to Previous EOS Version, Command: boot system flash , Description: If updating or rolling back the EOS version does not resolve the issue or introduces new problems, reverting to the previous version ensures the node returns to a known stable state.